



Society of Petroleum Engineers

SPE-229826-MS

Integrated Sedimentological Characterization to Identify Potential CCS Storage Site

Ana Widyanita, PETRONAS Research Sdn Bhd, Off Jalan, Ayer Itam, Kasawan Institusi Bangi, Kajang, Selangor, Malaysia; Uisdean Nicholson and Iain de Jonge-Anderson, Institute of GeoEnergy Engineering, School of Energy, Geoscience, Infrastructure & Society, Heriot-Watt University, Edinburgh, UK; Andreas Busch, The Lyell Centre, Heriot-Watt University, Edinburgh, UK

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229826-MS](https://doi.org/10.2118/229826-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

The development of hydrocarbon fields with high CO₂ content requires the integration of Carbon Capture and Storage (CCS) technologies, in alignment with national regulatory frameworks in Malaysia that prohibit CO₂ venting as part of greenhouse gas reduction strategies. Within offshore Peninsular Malaysia (PM), multiple geological storage options have been identified, including depleted hydrocarbon fields and deep saline aquifers. However, preliminary storage capacity assessments indicate that depleted fields alone are insufficient to accommodate the projected volumes of CO₂ from current and future high-CO₂ field developments. Consequently, saline aquifers represent a critical additional storage option, offering the potential to expand long-term capacity for large-scale CCS implementation.

The primary objective of this study is to assess the potential of saline aquifers within the Malay Basin as secure geological storage sites for CO₂. This assessment involves a detailed evaluation of core descriptions, facies groupings, depositional environment distributions, and the associated sealing and storage properties characterization. These analyses are essential for delineating the continuity, heterogeneity, and regional distribution of seal and storage intervals. The findings are intended to provide a systematic framework for evaluating the feasibility of long-term CO₂ containment in saline aquifers, thereby supporting regional-scale CCS deployment in the Malay Basin. A comprehensive basin-scale investigation was undertaken to identify prospective aquifer systems and define potential seal and storage intervals within the stratigraphic succession.

Regionally, the stratigraphic framework from Groups M/N/O to A/B records a progressive depositional shift from dominantly non-marine environments, including upper and lower coastal plain settings during the Oligocene synrift phase, towards increasingly tidal, estuarine, and marine environments through the Miocene to Pliocene. Within Group E, the central Malay Basin hosts multiple storage-prone intervals characterized by significant thickness, lower VClay values, and relatively higher porosity, deposited in a range of lower coastal plain, lacustrine, estuarine, and tidal settings. In contrast, the northwestern and eastern

parts of the basin are characterized by various thickness successions with higher VClay and lower porosity, indicating enhanced sealing potential in these regions.

The identification of saline aquifers with both storage and sealing capacity in the Malay Basin provides an opportunity to establish a regional CCS hub that could serve as a benchmark for future projects worldwide. Unlocking these storage sites not only enables the continued development of high-CO₂ hydrocarbon resources but also contributes to Malaysia's aspiration of achieving net-zero carbon emissions aspiration. Beyond national benefits, this study highlights technological and methodological advancements that may serve as a global reference for optimizing CCS deployment, offering a scalable framework for both regional and international applications.

Introduction

In Malaysia, the availability of hydrocarbon (HC) fields with low contaminant levels is progressively declining, while a significant number of remaining fields are characterized by elevated CO₂ content. The development of these high-CO₂ fields requires the implementation of Carbon Capture and Storage (CCS), in accordance with national regulatory frameworks that prohibit CO₂ venting. In addition to regulatory compliance, CCS plays a pivotal role in reducing greenhouse gas emissions and supports broader national and global climate change mitigation objectives. Within offshore Peninsular Malaysia (PM), several potential geological storage sites have been identified, including depleted hydrocarbon fields and deep saline aquifers (Figure 1). However, preliminary capacity assessments suggest that depleted fields alone are insufficient to accommodate the projected CO₂ volumes associated with current and future fields developments. Therefore, the inclusion of deep saline aquifers is essential to meet the long-term storage capacity demands, both for project-specific applications and for broader regional CCS deployment.

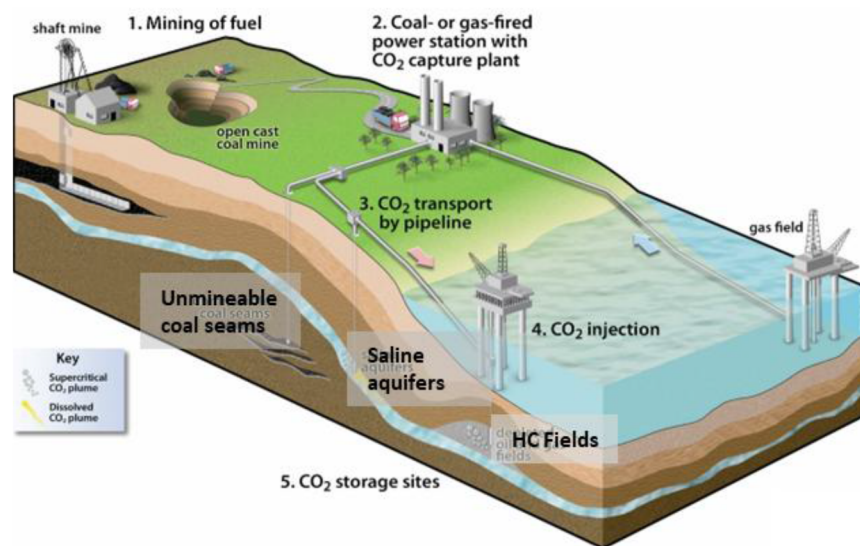


Figure 1—Current available storage site in Malaysia, which include depleted HC fields and saline aquifers

The identification and characterization of suitable regional saline aquifers for CCS in the Malay Basin pose substantial technical and geological challenges. Unlike hydrocarbon reservoirs, which have proven their capacity for long-term HC entrapment, saline aquifers have not undergone such validation, and their sealing integrity remains uncertain. A major limitation is the current lack of detailed understanding of the distribution, lateral continuity, and petrophysical properties of these deep aquifer systems. In this context, seal characterization becomes a critical component of CCS site evaluation. Comprehensive analysis of caprock integrity, including assessments of lithological continuity, mineralogical composition, pore structure, and sealing capacity, is essential to minimize leakage risks and ensure long-term containment of

injected CO₂. Furthermore, understanding the stratigraphic and structural relationships between potential reservoirs and their overlying seals is imperative for accurately assessing the suitability of regional saline aquifers as viable and secure CO₂ storage sites. This evaluation is further supported by an integrated understanding of sedimentology and the regional depositional environments, which play a fundamental role in controlling the distribution, architecture, and petrophysical properties of both reservoirs and sealing formations.

Objectives

The primary objective of this study is to conduct a detailed assessment of core descriptions, facies groupings, depositional environments, and the associated storage and sealing capacities of potential CO₂ storage sites within the Malay Basin. This is achieved through an integrated sedimentological and petrophysical approach, covering core-based sedimentary facies analysis, interpretation of depositional settings, and the evaluation of well-logs derived properties such as interval thickness, volume of shale (V_{shale}), and porosity. These analyses are essential for delineating the continuity and regional distribution of both seal and storage intervals. The integration of sedimentological and petrophysical data supports the evaluation of seal integrity and containment effectiveness, particularly with respect to the capacity of caprock units to retain injected CO₂ within underlying storage formations.

This research addresses several key scientific questions, including: (1) the identification of suitable regional saline aquifers as viable CO₂ storage sites within the basin; (2) the characterization and distribution of these aquifer systems; and (3) the evaluation of which depositional environments are most conducive to effective sealing and high-capacity storage. The findings aim to advance a robust methodology for assessing the feasibility of long-term CO₂ containment in saline aquifers, ultimately supporting regional-scale CCS deployment in the Malay Basin.

Regional Geology

There are two major basins in offshore Peninsular Malaysia (PM), which are the Malay Basin and the Penyu Basin ([Figure 2](#)). These extensional basins are in the southern part of the Gulf of Thailand and the South China Sea, positioned between mainland Southeast Asia and the major islands of Sumatra, Java, and Borneo. They form part of the continental core of Southeast Asia and are defined by a complex tectonostratigraphic evolution, comprising an initial phase of crustal extension associated with shear-related rift systems, followed by thermal subsidence, and subsequently, compressional deformation events ([Iftikhar et al., 2015](#); [Madon et al., 2020](#); [de Jonge-Anderson et al., 2024](#))

The Malay Basin is bounded by the Tenggol Arch and Penyu Basin to the southwest, the Natuna High and Pattani Basin to the northwest, the Khorat Swell and Nam Con Son Basin to the northeast, and the West Natuna Basin to the southeast ([Madon & Watts, 1998](#); [Madon et al., 2019](#)). This study targets the identification and evaluation of saline aquifers as potential carbon storage sites within the Malay Basin, which covers an area of approximately 50,000 km² and contains sedimentary successions thickness of up to 15 km ([Tjia, 1996](#); [Bishop, 2002](#)). Stratigraphically, the basin hosts sedimentary sequences from Group M to Group A, with hydrocarbon accumulations predominantly occurring in sandstone reservoirs of Groups K to D.

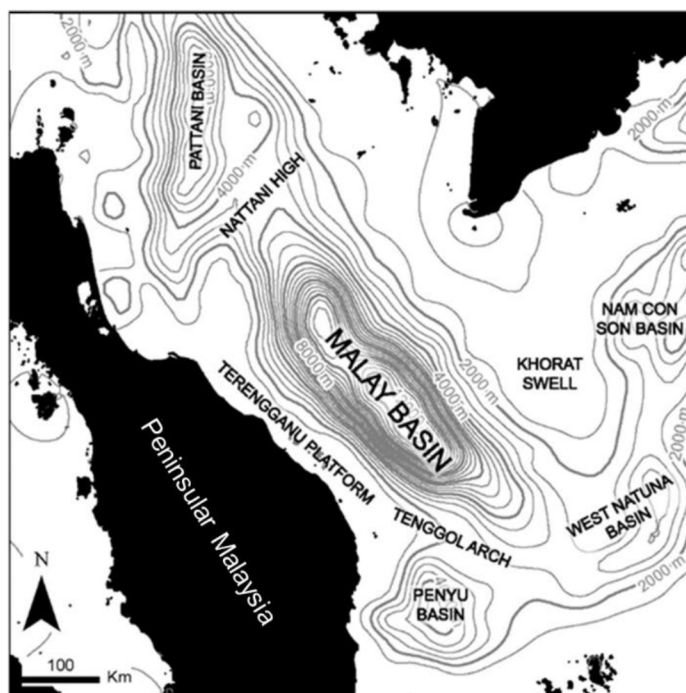


Figure 2—The Malay Basin total sediment thickness map and locations of major basins within offshore Peninsular Malaysia and Main-land Southeast Asia (Straume et al., 2019; de-Jonge Anderson, et al, 2024)

The stratigraphic evolution of the Malay Basin reflects the influence of three major tectonic phases. The first phase, the synrift stage, is represented by pre-Miocene rifting from Group M to Group K (Figure 3). During the Oligocene, sediment progradation occurred from Group M to Group L, with non-marine deposits transitioning from alluvial to coastal plain environments (Madon et al., 1999). Reservoir units within Group K are predominantly composed of deltaic and braided fluvial sediments (Madon et al., 1999). The second phase, the post-extensional stage of the Middle Miocene, covers Groups J to I and is characterized by thermal subsidence accompanied by tectonic subsidence. Sedimentation during this period reflects a transition from progradational to aggradational stacking patterns within fluvial–deltaic to tidally dominated estuarine depositional systems (Madon et al., 1999). The third phase, post-thermal subsidence from the Late Miocene to the present, is marked by regional compression affecting Groups H to A. Groups H and F were deposited during a transgressive phase associated with sea-level rise, comprising marine to deltaic facies. Groups E and D are dominated by tidal, shoreface, deltaic, and lower coastal plain deposits, with localized erosional unconformities. The youngest stratigraphic units, Groups B and A, consist primarily of transgressive marine silts and shales deposited nearshore to shallow marine environments.

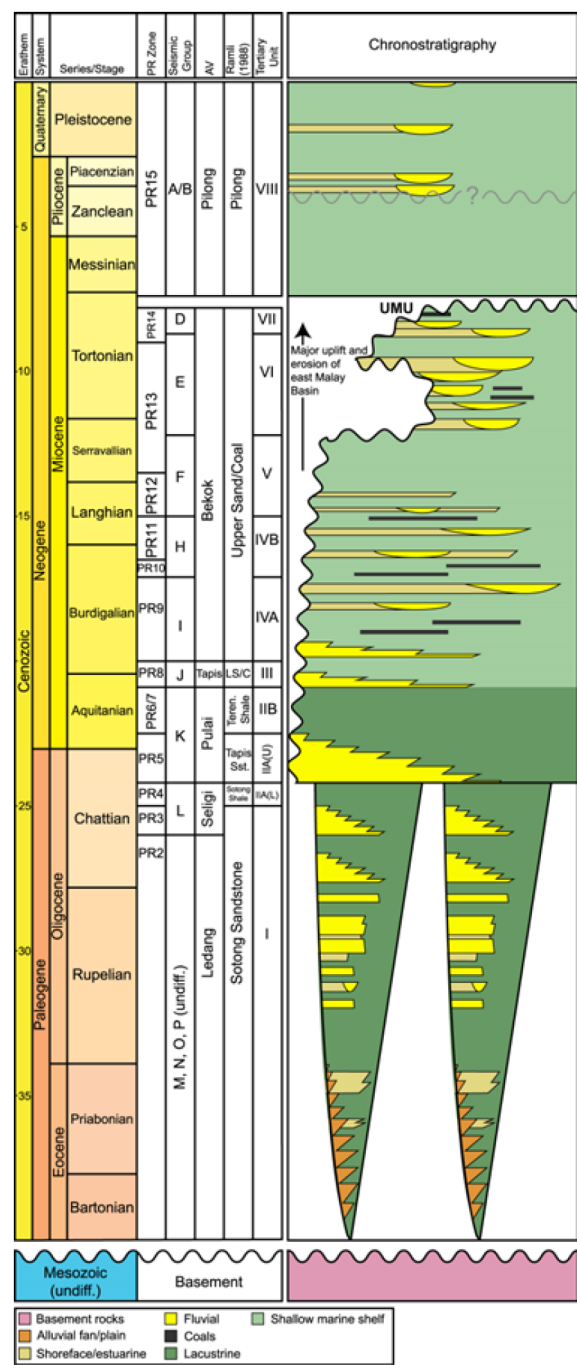


Figure 3—Malay Basin Stratigraphic chart which shows study intervals from deeper to shallower Group Sequence (Mansor et al., 2014; de-Jonge Anderson et al, 2024)

Data

The dataset for this study covers an area of approximately 50,000 km² within the Malay Basin, incorporating more than 100 identified fields. The study area is subdivided into three clusters (Figure 4): Cluster 1, located adjacent to the onshore region in the southern sector of the basin; Cluster 2, positioned in the northern region proximal to a group of high-CO₂ fields; and Cluster 3, situated furthest offshore in the eastern part of the basin. The investigation is supported by an extensive database comprising more than 3,000 wells and cumulative core lengths exceeding 15,000 m. Despite this, comprehensive field review reports and detailed evaluations of seal and storage characteristics are available for only a limited number of fields. Furthermore,

complete well log datasets are predominantly available for exploration and appraisal wells, necessitating the selection of key representative wells for seal and storage interval characterization.

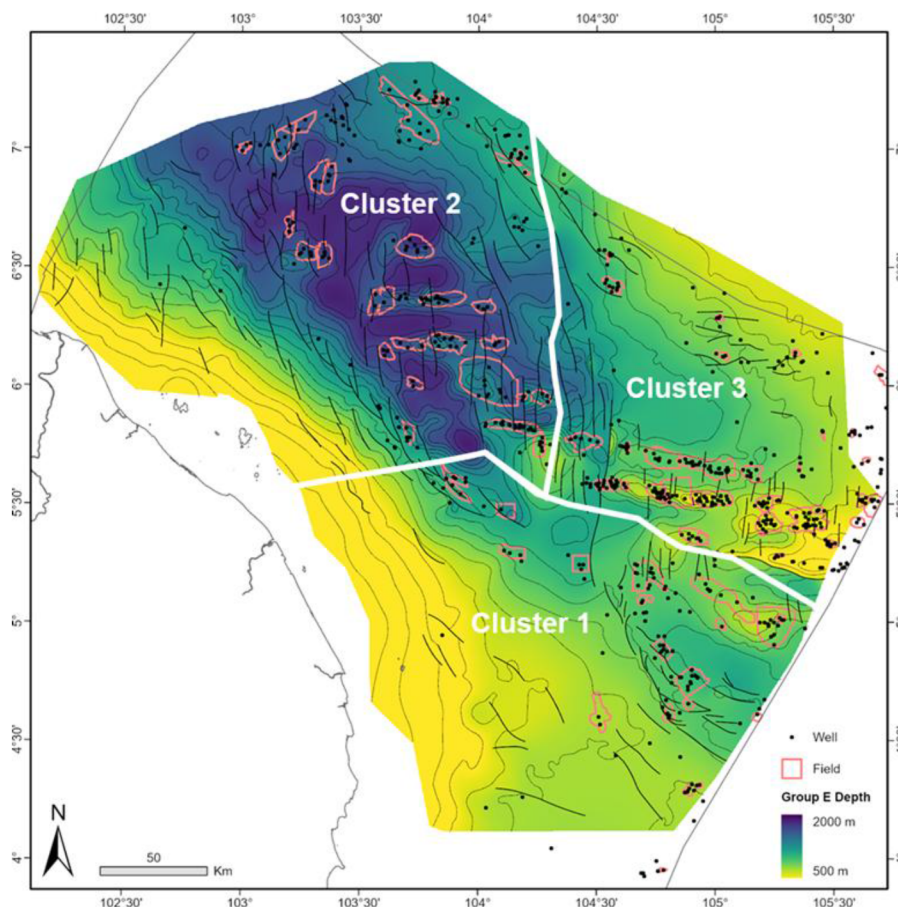


Figure 4—The Malay Basin Group E structure map with field polygons in pink and wells in black dots. The study area is divided into 3 clusters, which are cluster 1 (near onshore), cluster 2 (near high CO₂ fields cluster) and cluster 3 (furthest from onshore area)

The core dataset used in this study integrates multiple sources, including well logs, geological reports, final well reports, biostratigraphic analyses, sedimentological interpretations, Routine Core Analysis (RCA), and Special Core Analysis (SCAL). Together, these datasets enable the assessment of lithological, petrophysical, and depositional attributes essential for evaluating the suitability of regional saline aquifers as potential CO₂ storage sites.

Methodology

Comprehensive study was undertaken to evaluate the potential distribution of saline aquifers within the Malay Basin and identify prospective seal and storage reservoirs. This integrates regional geological framework analysis, core descriptions, facies classification, depositional environment mapping, and petrophysical property evaluation (Figure 5). A basin-wide review of existing fields was conducted across all stratigraphic group sequences to examine the occurrence and variability of potential storage and seal intervals in different depositional settings. Well-top selection was guided by detailed biostratigraphy interpretation, incorporating stratigraphic thinning and thickening trends, identification of transgressive and regressive sequences, and recognition of multiple stacked sand–shale successions within each stratigraphic group. Field-scale stratigraphic and depositional interpretations were then used to assess and characterize

the seals and storage intervals. Sand-dominated units were classified as potential storage intervals, whereas overlying shale-rich successions were designated as potential seals.

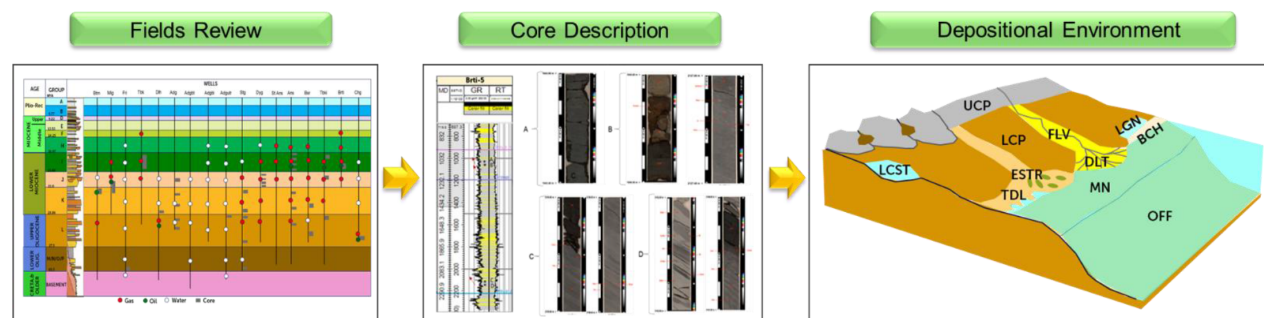


Figure 5—Methodology of the study, which covers the fields evaluation review, core sedimentological description and depositional environment interpretation

The delineation of these intervals relied on the integration of multiple datasets, including core sedimentological descriptions, lithological classifications, facies groupings, and depositional environment interpretations. Core description analysis included sedimentary structures, vertical and lateral sequence architecture, grain size distribution, texture, sorting, rounding, color variations, and bioturbation intensity. These sedimentological observations formed the basis for facies definition, which in turn supported depositional environment interpretations. The link between facies-depositional environment relationships and petrophysical attributes, such as interval thickness, clay volume (VClay), and porosity, was systematically evaluated to assess reservoir quality and seal effectiveness. These analyses underpin the identification of potential seal and storage pairs suitable for saline aquifer CO₂ sequestration within the Malay Basin.

Results and Discussion

For well Tjh-2 (Figure 6), four core intervals have been identified within the Group E stratigraphic sequence, designated as intervals 1, 2, 3, and 4. Core Interval 1 (1920–1921 m) comprises fine-grained sandstones characterized by ripple lamination, with symmetrical forms and unidirectional current influence. Occasional clay-rich (argillaceous) laminae and dispersed organic material are present. Vertical burrow traces occur but bioturbation intensity is generally low. A thin coal layer caps the interval. Gamma-ray (GR) log signatures display multiple serrated, coarsening-upward successions with a funnel-shaped motif. Integration of core sedimentology, facies classification, and well log patterns suggest deposition in a Lower Coastal Plain (LCP) to lacustrine (LCST) environment.

Core Interval 2 (1922–1923 m) consists of very fine-grained rippled sandstones, interbedded with siltstone and mudstone laminae. Carbonaceous material is concentrated at the base of the interval. Core Interval 3 (1924–1924.5 m) is dominated by mudstone containing very thin siltstone laminae, very fine grained, organic-rich, and exhibits small-scale load and slump structures, with localized siderite cementation. Core Interval 4 (1934–1935 m) comprises grey to dark-grey mudstones, parallel laminated with siltstone. The succession is organic-rich, very low bioturbation, and contains minimal sedimentary structure variation. GR log responses for intervals 2, 3, and 4 reveal multiple serrated, coarsening-upward successions with funnel-shaped motifs, consistent with lacustrine (LCST) depositional conditions

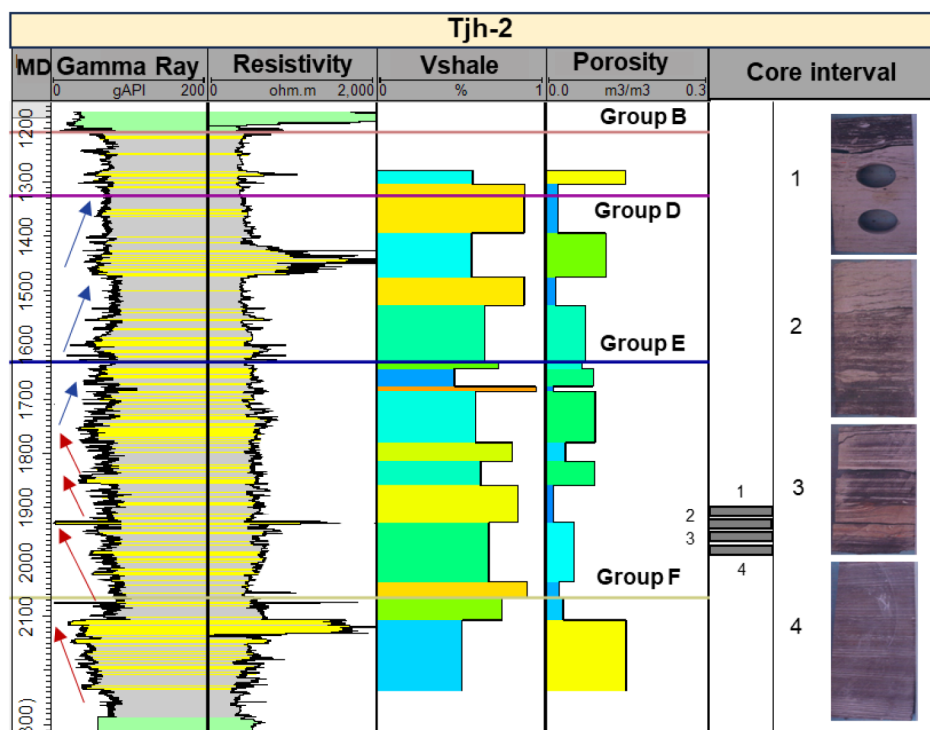


Figure 6—Tjh-2 core description and well logs characteristics, shows various facies group and different depositional environment interpretation

Core descriptions, facies classification, and depositional environment analyses were performed for all clusters and stratigraphic Group sequences. Within Group E, most fields in Cluster 2 are interpreted as having been deposited in lower coastal plain (LCP) settings, with lacustrine (LCST), tidal (TDL), and estuarine (ESTR) influences mainly in the central portion of the cluster. Fluvial and deltaic deposits are also present in the northwestern and southeastern areas of Cluster 2. During the Early to Middle Miocene, Group E experienced thermal and tectonic subsidence, which were accompanied by compressional deformation. These tectonic events resulted in localized inversion of half-graben structures through fault reactivation, as well as significant uplift and erosion in the southeastern part of the basin (Madon, 1999).

The classification of sand intervals as potential CO₂ storage units and shale intervals as potential seals was undertaken based on integrated depositional environment interpretations from core sedimentology, facies groupings, and petrophysical properties derived from well logs. Key parameters used in this assessment included thickness, volume of clay (VClay), and porosity. The integration of stratigraphic framework and depositional environment interpretations enabled the delineation of distinct sand and shale successions within each Group. Distribution patterns of these successions were further refined using gamma-ray (GR) log responses, VClay thresholds, and porosity cut-off values, which collectively guided the classification of intervals as either storage or seal units. Sand intervals meeting the porosity and VClay criteria were identified as potential storage targets, while the overlying shale units were designated as potential seals.

In Group E, nine discrete zones have been defined (Table 1). Seal intervals comprise zones E, E2, E4, E6, and E8, which are characterized by VClay values greater than 0.78 and porosity values less than 0.05. Storage intervals include zones E1, E3, E5, and E7, defined by VClay values less than 0.62 and porosity values greater than 0.08.

Table 1—Tjh-2 potential seal and storage properties for detail potential seal and storage layers which have variation of thickness, VClay and porosity average values

Zone	Thickness (m)	VClay (fraction)	Porosity (fraction)	Potential Seal/Storage
E	12.7	0.78	0.05	Seal
E.1	32.5	0.46	0.09	Storage
E.2	9.3	0.94	0.01	Seal
E.3	94.1	0.58	0.09	Storage
E.4	34.8	0.80	0.04	Seal
E.5	44.1	0.62	0.09	Storage
E.6	68.5	0.84	0.01	Seal
E.7	109.2	0.61	0.08	Storage
E.8	29.2	0.89	0.03	Seal

According to the porosity and VClay data in Group E, reveals a consistent inverse relationship, where higher porosity correlates with lower VClay and lower porosity correlates with higher VClay (Figure 7). This trend is similarly observed across various depositional environments, including lower coastal plain to lacustrine (LCP-LCST) environments (Figure 7). Based on these correlations, empirical equations have been derived to quantify the relationship between porosity and VClay. These equations provide a predictive tool for estimating reservoir and seal properties in specific stratigraphic groups and depositional environments, particularly in areas with limited data or where wells are absent.

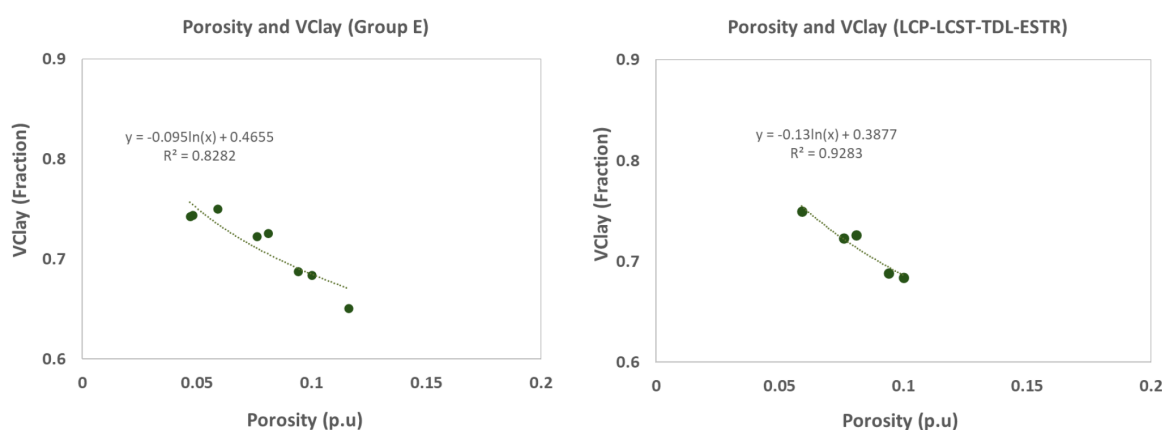


Figure 7—Group E Porosity and Vclay relationship (left) shows that the lower Vclay have higher porosity. Group E Lower Coastal Plain-Lacustrine-Tidal-Estuarine (LCP-LCST-TDL-ESTR) porosity and Vclay relationship (right) shows similar relationship for Group D sequences

Group E reaches a maximum thickness of approximately 700 m, with an average thickness of ~340 m (Figure 8). The greatest thicknesses occur in the central portion of Cluster 2, where deposition predominantly in lower coastal plain to lacustrine (LCP-LCST) environments. Thickness progressively decreases towards the northern areas of Cluster 2 and the southern regions of Clusters 1 and 3, where depositional settings transition to tidal (TDL) and estuarine (ESTR) environments. Fluvial, deltaic, estuarine, and tidal deposits generally exhibit thicknesses ranging from 50 to 350 m.

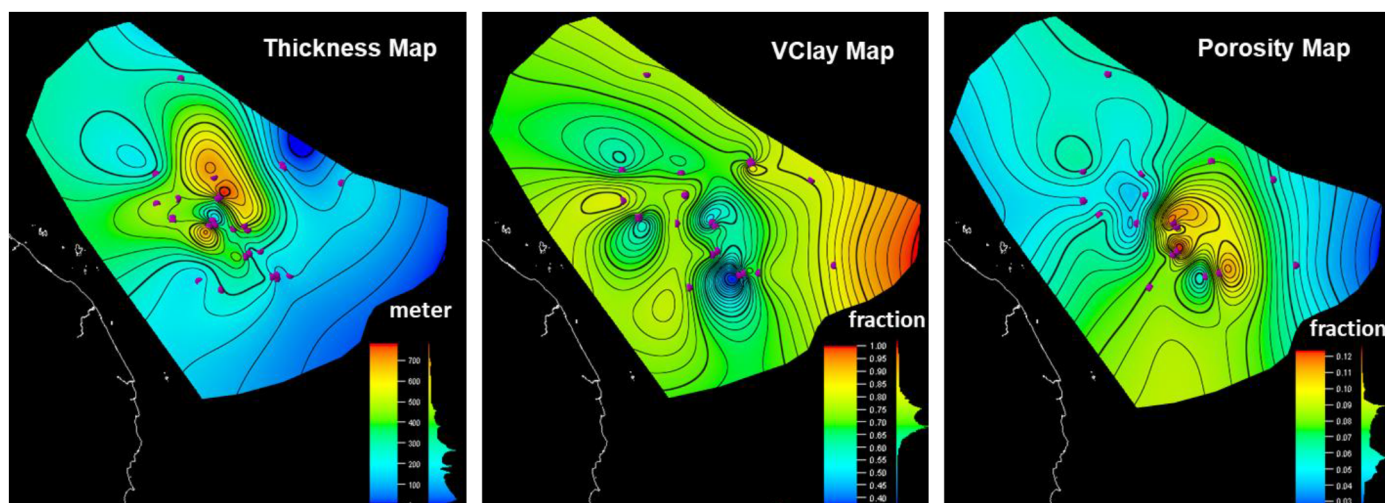


Figure 8—Thickness map (left), Vclay map (middle) and porosity map (right) of Group E. The potential seal area has the thickest interval, higher Vclay and lower porosity values. The potential storage area has the lower Vclay and higher porosity values

The VClay content within Group E varies between 0.46 and 0.84, with an average value of approximately 0.69 (Figure 8). Lower VClay values are observed in the central part of cluster 2 and the eastern area of cluster 1, where LCP-LCST environments predominate, indicating potential storage intervals. In contrast, Vclay values in areas characterized by fluvial, deltaic, estuarine, and tidal environments range from 0.6 to 0.85. This group shows potential for seals distributed throughout the northern area of cluster 2 and western area of cluster 1.

The porosity within Group E ranges from 0.04 to 0.18, with an average of approximately 0.08 (Figure 8). The highest porosity values in Group E are found in the southeastern part of cluster 2 and the northern area of cluster 1, where LCP-LCST environments predominant, indicating favorable conditions for storage. In contrast, porosity values in fluvial, deltaic, estuarine, and tidal settings range from 0.04 to 0.06. This group exhibits seal potential dispersed throughout the northern area of cluster 2.

Based on the integrated thickness, VClay, and porosity analyses, the northwestern and eastern parts of Cluster 2 present significant seal potential, linked to deposition in fluvial, deltaic, tidal, and estuarine environments. The central Malay Basin, dominated by LCP-LCST settings, is characterized by thicker sediment accumulation, lower VClay, and higher porosity, making it a prime target for Group E storage. During the Early to Middle Miocene, Group E was affected by thermal and tectonic subsidence, compressional deformation, and localized uplift and erosion in the southeastern basin, leading to the absence of Group E deposits in that area.

Conclusions

A comprehensive basin-scale assessment was undertaken to evaluate the distribution of saline aquifers and delineate potential seal and storage intervals within the stratigraphic framework. This analysis integrates regional geological evaluation, core description, facies classification, depositional environment mapping, and petrophysical property analysis. By combining stratigraphic group sequence analysis with depositional environment interpretations, sand-rich intervals were identified as potential storage units, while overlying shale-dominated intervals were classified as potential seals.

Depositional environment mapping across the Malay Basin revealed a range of settings, including upper coastal plain (UCP), lower coastal plain (LCP), fluvial (FLV), deltaic (DLT), lacustrine (LCST), estuarine (ESTR), tidal (TDL), and marine (MN) environments. In the overall stratigraphic sequence from Group M/N/O to A/B, there is a progressive shift from dominantly non-marine conditions, including UCP and

LCP settings during the Oligocene synrift phase, to the gradual occurrence of tidal, estuarine, and marine depositional environments from the Miocene to Pliocene periods.

In the central Malay Basin, Group E contains multiple storage-prone intervals, characterized by variable thickness, lower VClay, and higher porosity. These deposits formed in a range of depositional settings, including lower coastal plain, lacustrine, estuarine, and tidal environments. In contrast, the northwestern and eastern parts of the basin are characterized by various thickness of sediment packages, higher VClay values, and lower porosity, indicating greater sealing potential in these areas. Due to the tectonic subsidence and compressional deformation causing local uplift and erosion, resulting in the absence of Group E deposits in the southeastern region.

Nine zones have been identified for potential seal and storage intervals within this Group. Seal-prone units include zones E, E2, E4, E6, and E8, which are characterized by VClay values greater than 0.78 and porosity values less than 0.05. Storage-prone units include zones E1, E3, E5, and E7, defined by VClay values less than 0.62 and porosity values greater than 0.08. These results provide a detailed framework for identifying and mapping CO₂ storage and seal distributions in the Malay Basin.

References

- Bishop, M.G., 2002. Petroleum systems of the Malay Basin province, *Malaysia. U.S. Geological Survey Open-File Report 99-50T*. Department of the Interior, Denver. pg. 25
- De Jonge-Anderson, I., Widyanita, A., Busch, A., Doster, F., Nicholson, U., 2024. New insights into the structural and stratigraphic evolution of the Malay Basin using 3D seismic data: Implications for regional carbon capture and storage potential, *Basin Research*, **36**, pg. 2–24
- Hassan, M., Swapan, K., Manoj, J., Numair, A., 2017, Cenozoic development of southwestern Malay Basin; new insights from subsidence analysis and thermal history, *Arab Journal Geoscience*, **10**: 192, pg. 2–5
- Iftikhar, A., Deva, G., Wan, I., 2015, 3D Predrill Pore Pressure Prediction Using Basin Modelling Approach in a Field of Malay Basin, *Asian Journal of Earth Science*, **8**, pg. 24–31
- Madon, M., Watts, A., 1998. Gravity anomalies, subsidence history and the tectonic evolution of the Malay and Penyu Basin (offshore Peninsular Malaysia), *Basin research*, **10**, pg. 375–381
- Madon, M., Peter, A., Jamaal, H., Mansor, A., 1999. Malay Basin. In: *PETRONAS "The Petroleum Geology and Resources of Malaysia"*, Chapter 8, pg. 177–184
- Madon, M., John J., Franz L., Colm, M., 2019, Overview of the structural framework and hydrocarbon plays in the Penyu Basin, offshore Peninsular Malaysia, *Bulletin of the Geological Society of Malaysia*, Volume. **68**, pg 1–23
- Madon, M., John J., Franz L., Hafiz, D., Khairil, A., 2020, Pre-Tertiary basement subcrops beneath the Malay and Penyu basins, Offshore Peninsular Malaysia: Their recognition and hydrocarbon potential, *Bulletin of the Geological Society of Malaysia*, Volume. **70**, pg 163–193
- Madon, M., 2021, Five decades of petroleum exploration and discovery in the Malay Basin (1968–2018) and remaining potential, *Bulletin of the Geological Society of Malaysia*, Volume. **72**, pg 63–88
- Mansor, M. Y., Rahman, A. H. A., Menier, D., Pubellier, M., 2014, Structural evolution of Malay Basin, its link to Sunda Block tectonics. *Marine and Petroleum Geology*, **58**, 736–748
- Morley, C.K., 2004. Nested strike-slip duplexes, and other evidence for Late Cretaceous–Palaeogene transpressional tectonics before and during India–Eurasia collision, in Thailand, Myanmar and Malaysia. *Journal of the Geological Society of London*, **161**, pg. 799–812
- Peter Lunt, A reappraisal of the Cenozoic stratigraphy of the Malay and West Natuna Basin, *Journal of Asian Earth Science*, 2021, pg. 1–13
- Ramli, M. D. 1988. Stratigraphy and paleofacies development of Carigali's operating areas in the Malay Basin, South China Sea. *Bulletin of the Geological Society of Malaysia*, **22**, 153–187
- Straume, Em O., Gaina, C., Medvedev, S., Hochmuth, K., Gohl, K., Whittaker, J. M., Abudul Fattah, R., Doornenbal, J. C., Hopper, J. R., 2019. Globbed: Updated total sediment thickness in the world's oceans. *Geochemistry, Geophysics, Geosystems*, **20**(4), 1756–1772
- Tjia, H.D., 1994, Inversion tectonics in the Malay Basin: evidence and timing of events. *Bulletin Geological Society of Malaysia*, **36**, pg. 119–126
- Tjia, H.D., Liew, K.K., 1996, Changes in tectonic stress field in northern Sunda Shelf basins. In: Hall, R., Bulndell, D.J., eds., *Tectonic Evolution of Southeast Asia. Geological Society of London Special Publication*, **106**. Pg. 291–306